

# Loan Approval Prediction

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# Introduction

- In an era characterized by data-driven decision-making, financial institutions are constantly seeking ways to streamline their operations, enhance customer experiences, and mitigate risks
- Predicting loan approval is crucial for financial institutions.
- Our loan approval prediction model is poised to revolutionize the lending process by harnessing the power of advanced machine learning and predictive analytics.
- Introducing a Random Forest model with class imbalance handling.



# DATASET OVERVIEW

The Dataset contains 12 feautres

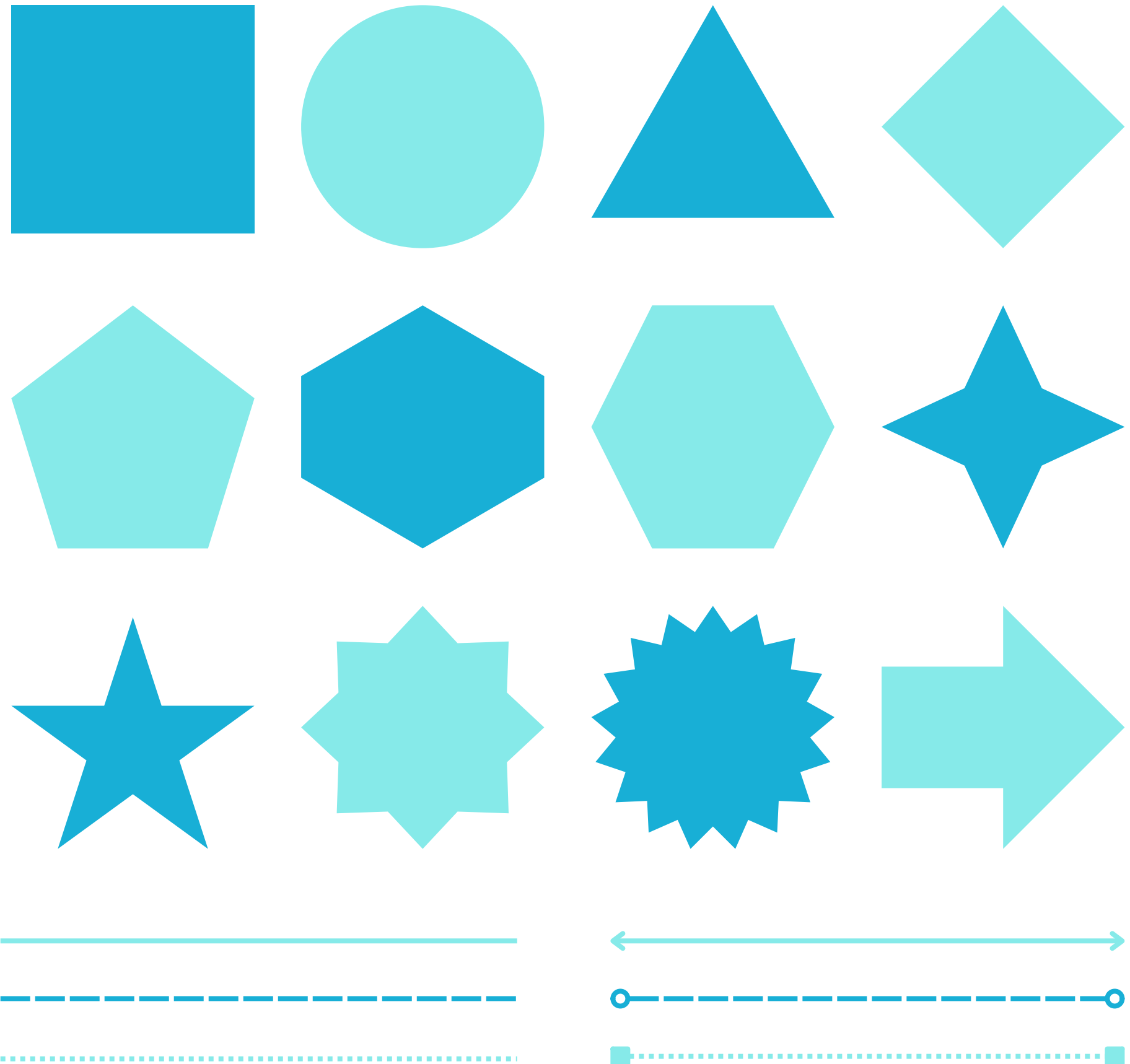
| no_of_dependents | education    | self_employed | income_annum | loan_amount | loan_term | cibil_score | residential_assets_value | commercial_assets_value | l  |
|------------------|--------------|---------------|--------------|-------------|-----------|-------------|--------------------------|-------------------------|----|
| 2                | Graduate     | No            | 9600000      | 29900000    | 12        | 778         | 2400000                  | 17600000                | 1  |
| 0                | Not Graduate | Yes           | 4100000      | 12200000    | 8         | 417         | 2700000                  | 2200000                 | 2  |
| 3                | Graduate     | No            | 9100000      | 29700000    | 20        | 506         | 7100000                  | 4500000                 | 3  |
| 3                | Graduate     | No            | 8200000      | 30700000    | 8         | 467         | 18200000                 | 3300000                 | 4  |
| 5                | Not Graduate | Yes           | 9800000      | 24200000    | 20        | 382         | 12400000                 | 8200000                 | 5  |
| ...              | ...          | ...           | ...          | ...         | ...       | ...         | ...                      | ...                     | 6  |
| 5                | Graduate     | Yes           | 1000000      | 2300000     | 12        | 317         | 2800000                  | 500000                  | 7  |
| 0                | Not Graduate | Yes           | 3300000      | 11300000    | 20        | 559         | 4200000                  | 2900000                 | 8  |
| 2                | Not Graduate | No            | 6500000      | 23900000    | 18        | 457         | 1200000                  | 12400000                | 9  |
| 1                | Not Graduate | No            | 4100000      | 12800000    | 8         | 780         | 8200000                  | 700000                  | 10 |
| 1                | Graduate     | No            | 9200000      | 29700000    | 10        | 607         | 17800000                 | 11800000                | 11 |

# DATA PREPROCESSING

- **Handling Missing Values:**
  - Utilized forward fill method to fill missing values.
  - Ensured dataset completeness and consistency.
- **Encoding Categorical Features:**
  - Employed one-hot encoding for categorical variables (education and self\_employed).
  - Transformed categorical data into numerical format suitable for modeling.
- **Feature Scaling:**
  - Applied Min-Max scaling to normalize numeric features.
  - Brought all features within the same scale to prevent bias towards specific attributes.

# Class Imbalance Handling

- In loan approval prediction, the dataset often contains an unequal distribution of loan approval classes.
- Imbalanced classes can lead to biased model performance and poor predictions.
- Techniques Used: SMOTE and Random Under-Sampling:
  1. SMOTE (Synthetic Minority Over-sampling Technique): Oversamples the minority class by generating synthetic examples, balances the class distribution, and reduces bias, enhancing the classifier's ability to generalize.
  2. Random Under-Sampling: Undersamples the majority class by randomly removing instances, addressing class imbalance by reducing the dominance of the majority class, and helping the model focus on minority class patterns.





## SMOTE VISUALIZATION

Bar plot illustrating the distribution of the target class before and after SMOTE.

# MODEL BUILDING



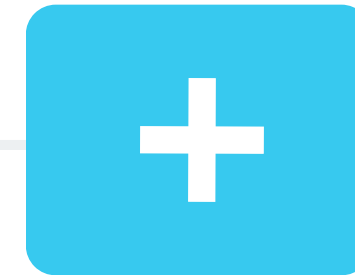
## RANDOM FOREST CLASSIFIER

Powerful ensemble learning algorithm, was chosen as the foundation of our loan approval prediction model. Its ability to handle complex relationships and feature interactions makes it well-suited for this task.



## CLASS IMBALANCE APPROACHES

To tackle the challenge of class imbalance, two distinct approaches were employed: using the original dataset and a version with under-sampled data. This allowed us to comprehensively evaluate the model's performance under different conditions.



## MODEL TRAINING

The model was trained using both the original and under-sampled datasets. This involved feeding the algorithm with features extracted from the data, allowing it to learn and identify patterns associated with loan approval.



## OBSERVATIONS

Achieved impressive performance on both original and under-sampled datasets.

# MODEL EVALUATION METRICS

Four different metrics were used

2

3

4

5

## ACCURACY

- Random Forest (Under-sampled): 0.9784
- Random Forest: 0.9784
- High accuracy indicates the overall correctness of predictions.

## PRECISION

- Random Forest (Under-sampled): 0.9669
- Random Forest: 0.9597
- Precision measures the ratio of correctly predicted positive instances among the predicted positives.

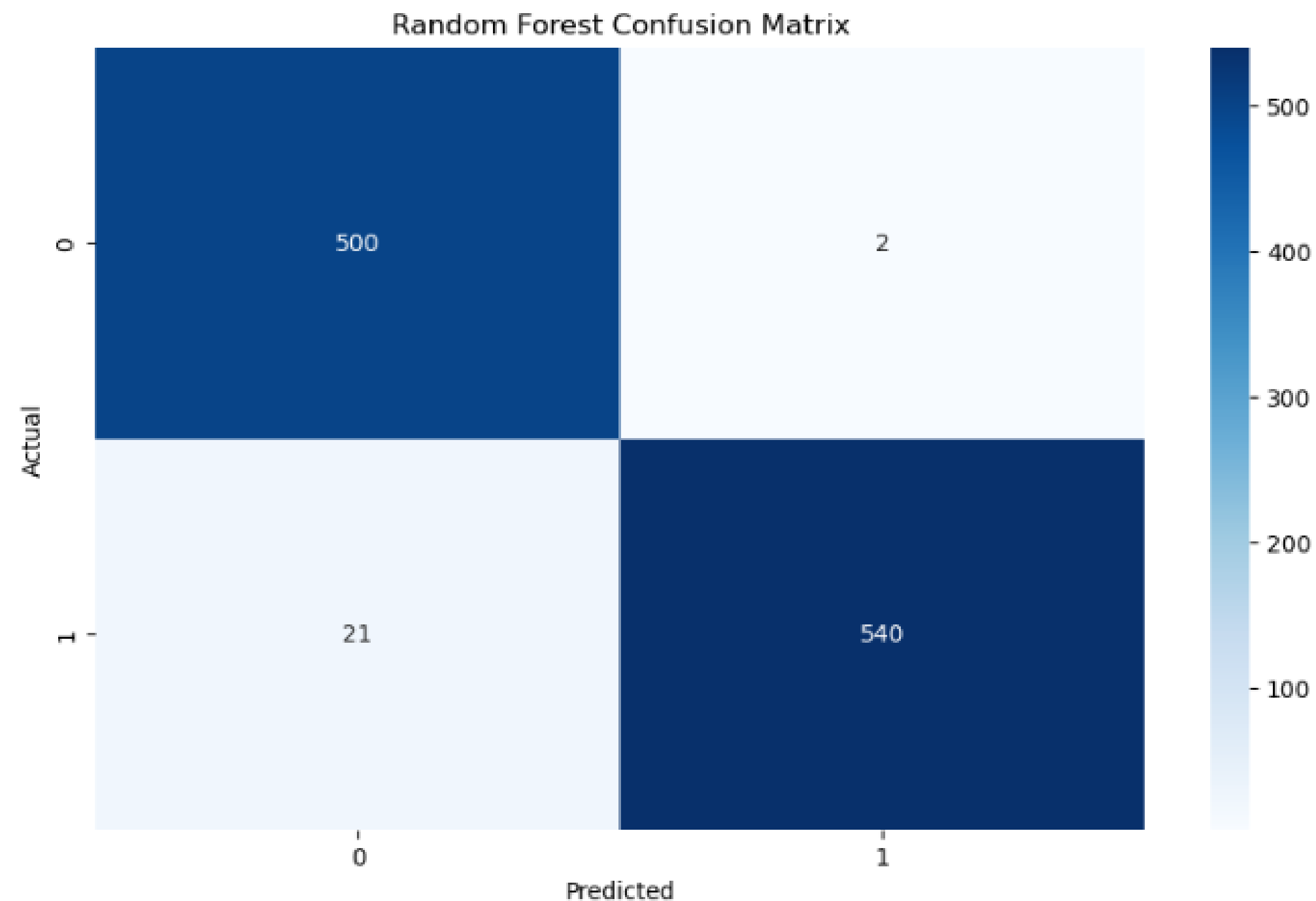
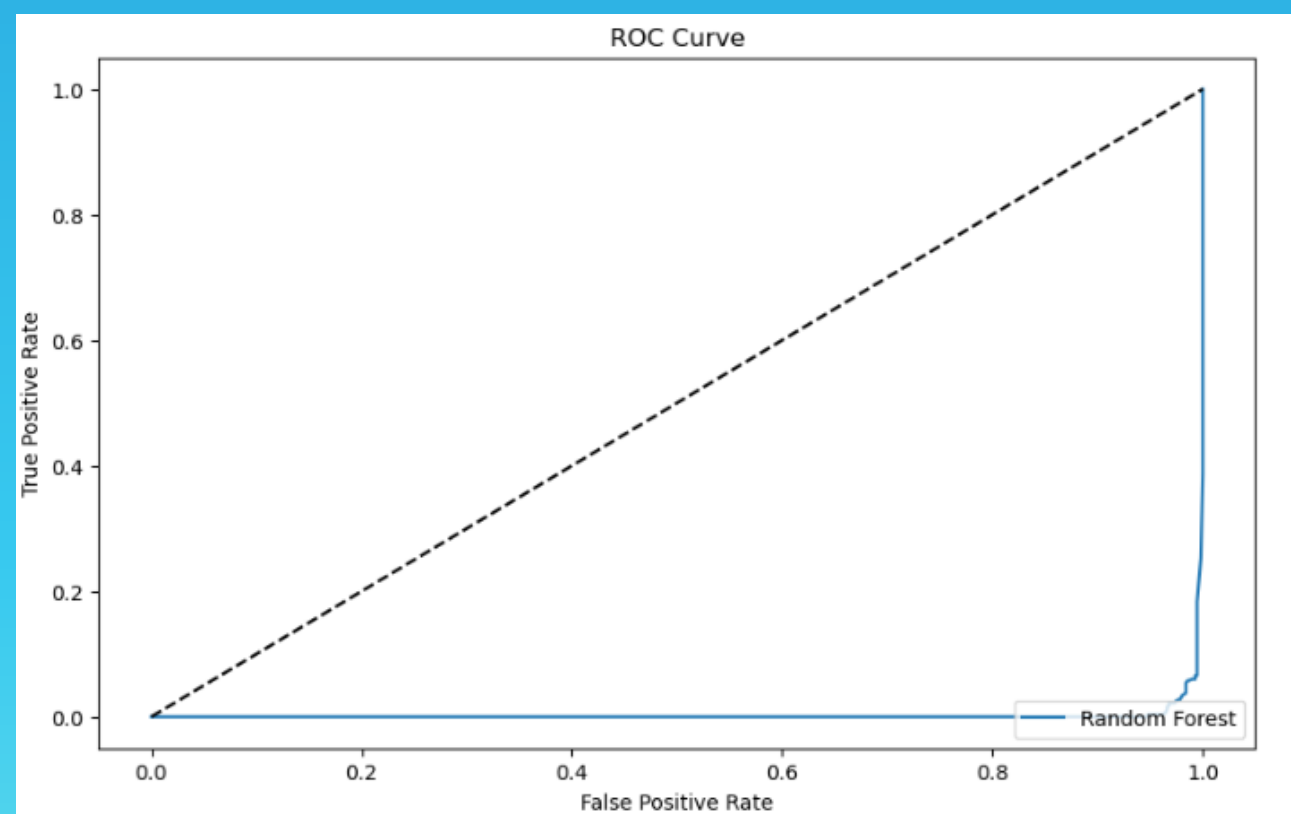
## RECALL

- Random Forest (Under-sampled): 0.9880
- Random Forest: 0.9960
- Recall represents the ratio of correctly predicted positive instances among all actual positives.

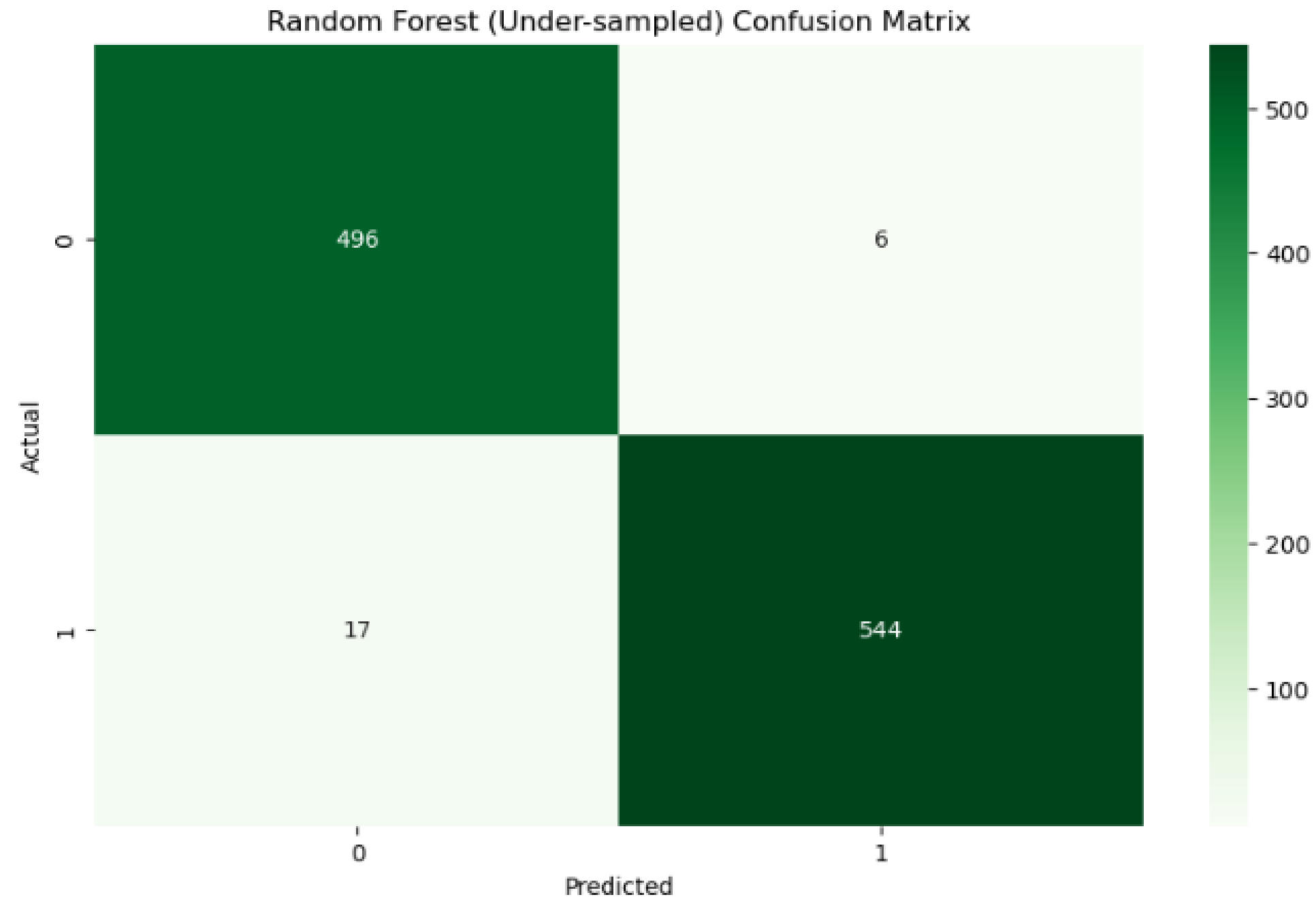
## F1 SCORE

- Random Forest (Under-sampled): 0.9773
- Random Forest: 0.9775
- F1 Score is the harmonic mean of precision and recall, providing a balanced measure of model performance.

# Model Performance - Original

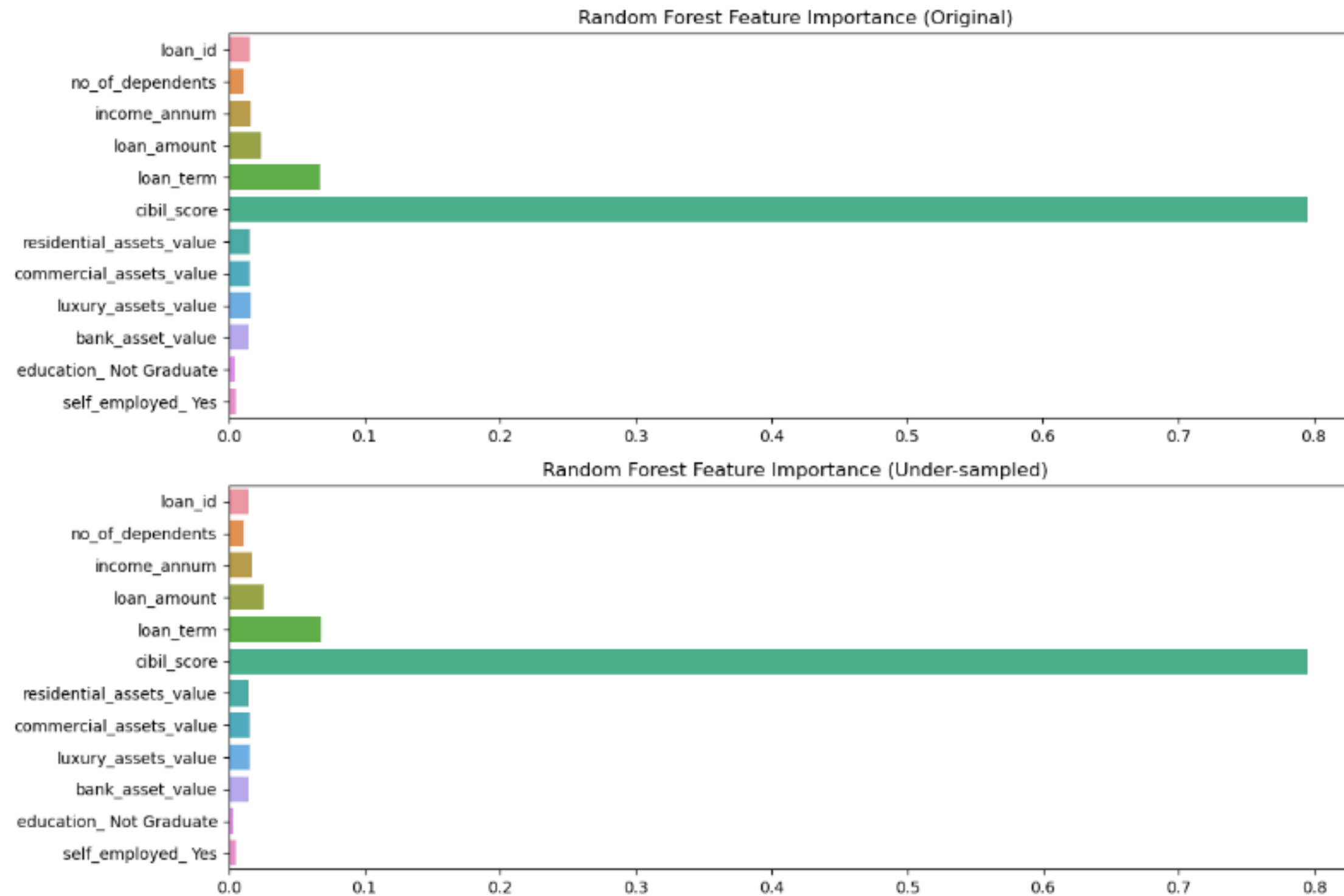


# Model Performance - Under- Sampled



# FEATURE IMPORTANCE

- Understanding the significance of input features in predicting loan approval.
- Identifying the features that contribute most to the model's decision-making process.



# CONCLUSION

1

Our loan approval prediction model utilizing the Random Forest algorithm demonstrated impressive performance.

2

Class imbalance techniques, including SMOTE and Random Under-Sampling, significantly improved model performance.

3

The achieved accuracy, precision, recall, and F1 Score demonstrate the model's robustness in handling class imbalance and predicting loan approvals effectively.

4

Our model's success underscores its potential application in real-world scenarios to enhance loan approval decision-making, contributing to more informed and accurate lending practices.

5

Further exploration and refinement of the model hold promising avenues for continued improvement, such as hyperparameter tuning and additional feature engineering

THANK  
YOU!

